

Appendix C: Earth-System Context for Hydraulic Collapse

Supplementary Material

This appendix provides supplementary analysis for the paper "Rapid Continental Reorganization Through Hydraulic Collapse: A Solution to the Heat Problem in Catastrophic Plate Tectonics" (Longmire, 2025). It outlines a broader geophysical environment in which hydrotectonic collapse could operate.

Purpose: The aim is not to introduce new required mechanisms but to situate the collapse within a plausible Earth-system response involving hydrology, atmospheric dynamics, and potential triggering events.

Relationship to core model: These considerations do not replace the core model and should be treated as conceptual extensions requiring further quantitative study. The primary heat solution (water-mediated friction dissipation at shallow depths) presented in the main paper stands independently of these atmospheric calculations.

Status: Order-of-magnitude estimates with acknowledged uncertainties. Demonstrates plausibility and provides bounds rather than precise predictions.

C.1. Potential Triggering Environments

The hydrotectonic model does not depend on any single trigger. Basin seals under high pore pressure can fail spontaneously once stress thresholds are exceeded. However, several geophysical processes could increase the likelihood of synchronous or near-synchronous failure across large regions.

Possible contributors include:

1. **Supervolcanic eruptions (VEI-7/VEI-8).**

These events generate strong seismic waves, crustal flexure, and extensive fracturing that can perturb pore pressure regimes. Their thermal effects are primarily vertical and localized rather than laterally transmitted.

2. Large impacts (including Chicxulub-scale).

Impacts produce regional pressure redistribution, seismic shaking, and transient permeability increases. The thermal pulse vents upward and dissipates primarily through atmospheric injection. Their role here is mechanical, not thermal.

3. Progressive overpressure within sedimentary basins.

Large, sealed basins accumulating fluids over time represent well-documented metastable systems. Once critical pressures are reached, they can initiate basin-scale collapse independent of external forcing.

These processes should be viewed as *possible contributors*, not essential components, and all require further modeling to quantify their potential effect on basin-scale pore-pressure thresholds.

C.2. Hydrological Dynamics Under Global Inundation

If large regions became rapidly inundated, hydrological processes would dominate surface and near-surface heat transport. Evaporation, condensation, and precipitation represent efficient mechanisms for redistributing energy within the atmosphere-ocean system.

C.2.1. Evaporative Heat Export

When ocean surface temperatures exceed equilibrium (due to energy input from crustal processes), evaporation rate increases substantially. The latent heat of vaporization removes energy from the ocean without raising atmospheric temperature until condensation occurs at altitude.

Order-of-magnitude estimate:

Assumptions:

- Ocean surface area: $A_{\text{ocean}} \approx 5 \times 10^{14} \text{ m}^2$ (roughly 1.4× modern ocean area during peak inundation)
- Excess surface temperature: $\Delta T \approx 5\text{-}10 \text{ K}$ above equilibrium
- Enhanced evaporation rate: $E \approx 10\text{-}20 \text{ mm/day}$ (compared to modern $\sim 3 \text{ mm/day}$ average)
- Latent heat of vaporization: $L_v \approx 2.5 \times 10^6 \text{ J/kg}$
- Water density: $\rho = 1000 \text{ kg/m}^3$

Heat flux calculation:

$$Q_{\text{evap}} = A_{\text{ocean}} \times E \times \rho \times L_v$$

Using $E = 15 \text{ mm/day} = 1.74 \times 10^{-7} \text{ m/s}$:

$$Q_{\text{evap}} \approx (5 \times 10^{14} \text{ m}^2) \times (1.74 \times 10^{-7} \text{ m/s}) \times (10^3 \text{ kg/m}^3) \times (2.5 \times 10^6 \text{ J/kg})$$

$$Q_{\text{evap}} \approx 2.2 \times 10^{17} \text{ W}$$

Per-unit-area heat flux:

$$q_{\text{evap}} \approx 430 \text{ W/m}^2$$

This represents approximately 60× the heat flux from friction dissipation calculated in the main paper (~7 W/m² sustained water-film dissipation). However, this is an upper bound under peak thermal stress conditions. The actual sustained rate would be lower and depends on atmospheric capacity to export the latent heat.

C.2.2. Atmospheric Radiative Export

The critical question is whether evaporated water can be efficiently radiated to space or whether energy accumulates in the atmosphere, creating thermal runaway.

Key factors preventing accumulation:

1. **High-altitude condensation:** Storm systems transport water vapor to the upper troposphere (8-12 km) and potentially lower stratosphere (12-20 km). At these altitudes, radiative cooling is far more efficient than at sea level.
2. **Radiative efficiency scaling:** Outgoing longwave radiation scales approximately as T^4 (Stefan-Boltzmann). Even modest temperature increases at high altitude substantially increase energy export.

Order-of-magnitude check:

Modern Earth radiates ~240 W/m² to space on average. If atmospheric temperature at radiating altitude increases by ~10%, outgoing radiation increases by:

$$\Delta Q_{\text{rad}} \approx 4 \times Q_{\text{rad}} \times (\Delta T/T) \approx 4 \times 240 \times 0.1 \approx 96 \text{ W/m}^2 \text{ additional export}$$

This provides substantial margin for increased heat export capacity during peak thermal stress.

Caveat: These calculations assume the atmosphere can maintain radiative balance. If heat input exceeds export capacity for extended periods, temperatures could rise beyond these estimates. The coupling between evaporation rate, condensation altitude, and radiative efficiency requires detailed climate modeling to verify.

C.3. Hypercane-Class Storms as Potential Heat Transporters

Theoretical work on hypercanes suggests that extremely warm ocean surfaces can generate storms with very high latent heat fluxes. In a globally inundated environment with anomalously warm surface temperatures, such storms could form more readily and may represent a significant atmospheric heat-removal pathway.

C.3.1. Formation Conditions

Under conditions where localized ocean regions reach 305-315 K (32-42°C) due to proximity to active collapse zones, atmospheric instability would drive formation of intense tropical cyclone analogs. While full hypercane conditions (requiring >323 K or 50°C) may not be reached, exceptionally intense hurricanes operating continuously across warm water regions become plausible.

C.3.2. Heat Transport Capacity

Order-of-magnitude estimate:

A modern Category 5 hurricane dissipates $\sim 6 \times 10^{14}$ W at peak intensity. Enhanced storm systems operating continuously over warm water could sustain higher dissipation rates. If multiple such systems operate simultaneously across different ocean basins, total atmospheric heat transport could reach:

$$Q_{\text{storm}} \sim 10^{15} - 10^{16} \text{ W}$$

Mechanism: Storm updrafts transport latent heat from surface to altitude (10-15 km), where:

- Condensation releases heat into upper troposphere
- Radiative cooling to space becomes highly efficient
- Heat is exported rather than accumulating at surface

Critical distinction: Storm systems act as convective heat pumps, moving energy from the region where it threatens the biosphere (surface) to regions where it can be radiated to space (high altitude). This redistributes heat already introduced into the system rather than generating additional heat.

C.3.3. Uncertainties

A complete quantitative assessment would require:

- Ocean surface temperature evolution modeling
- Latent heat flux models for hypercane-strength systems
- Radiative transfer modeling for high-altitude condensation
- Storm lifetime and frequency predictions
- Coupled ocean-atmosphere simulations

These calculations are beyond the scope of this appendix but represent important directions for strengthening the atmospheric heat export component of the model.

C.4. Integration with Primary Heat Budget

The atmospheric mechanisms described above operate in parallel with the water-mediated friction dissipation calculated in the main paper. They do not replace the primary mechanism but provide additional heat export capacity during periods of peak thermal stress.

Heat budget summary:

1. **Primary source:** Gravitational potential energy release during crustal collapse ($\sim 10^{23}$ J total)
2. **Primary sink:** Water-film dissipation at shallow depths (~ 7 W/m² sustained)
3. **Secondary sink:** Evaporative heat export (~ 100 - 400 W/m² during peak thermal stress, declining as ocean cools)
4. **Tertiary sink:** Storm-driven atmospheric export ($\sim 10^{15}$ - 10^{16} W during active storm periods)

Temporal dynamics:

- **Early phase (days to weeks):** High evaporative and storm activity as ocean surface warms from crustal energy input. Peak atmospheric heat export.
- **Middle phase (weeks to months):** Declining evaporative flux as ocean approaches new equilibrium. Storm activity moderates.

- **Late phase (months):** Evaporative flux returns toward normal. Atmospheric heat export declines to near-modern levels.

Key result: The atmospheric pathways provide a substantial buffer during the period of peak thermal stress. Even if the water-film dissipation calculation in the main paper is conservative by a factor of 2-3, the atmospheric mechanisms provide additional margin preventing thermal runaway.

Important caveat: These atmospheric calculations are order-of-magnitude estimates with significant uncertainties (see Section C.8). They demonstrate plausibility and provide bounds rather than precise predictions. The core heat solution remains the shallow water-mediated friction collapse presented in the main paper.

C.5. Interaction Between Surface Hydrology and Crustal Failure

Hydrologically dominated environments have several consequences for crustal deformation:

1. **Enhanced fluid circulation** into dilatant shear zones may improve advective heat removal beyond the conservative estimates in the main paper.
2. **Increased precipitation and surface loading** can alter shallow stress fields and pore pressures, potentially affecting the timing and distribution of basin failures.
3. **High water availability** can accelerate the formation of fluid-rich damage zones during deformation, promoting rapid failure propagation.

These effects do not change the basic hydraulic collapse mechanism but suggest ways in which surface hydrology could interact with the evolving crust during the event. Quantifying these interactions would require coupled crust-hydrosphere simulations.

C.6. Observational Versus Global-Scale Perspective

An observer located far from triggering events (basin ruptures, volcanic centers, impact sites) would experience a hydrologically dominated environment rather than direct exposure to local catastrophic processes. From a local perspective, the system would manifest primarily as:

- Persistent rainfall from enhanced atmospheric circulation
- Rising water levels from basin discharge and ocean reorganization
- Periodic turbulence from long-period waves and storm systems
- Extended darkness from atmospheric aerosol loading

Meanwhile, at the global scale:

- Crustal blocks undergoing episodic motion via hydraulic collapse
- Basin-scale subsidence creating new accommodation space
- Distributed failure propagating through connected overpressured domains
- Ocean reorganization as basins deepen and continental regions uplift

This distinction between local experience and global geophysical processes is included for interpretive clarity. The local observer experiences the hydrological and atmospheric manifestations of processes operating at crustal and basin scales.

C.7. Testable Aspects

While direct observation of the proposed event is impossible, several aspects of this Earth-system framework are testable:

Modern analogs for atmospheric heat export:

- Study heat export efficiency in regions with anomalously warm ocean surfaces (e.g., volcanic islands, hydrothermal ocean regions)
- Validate evaporative flux scaling and storm response under extreme thermal gradients
- Test coupled ocean-atmosphere models under globally flooded boundary conditions

Paleoclimate proxies (if this event occurred):

- Globally distributed storm deposits at a single stratigraphic horizon
- Isotopic signatures in carbonates indicating sustained high evaporation rates
- Anomalous atmospheric composition proxies (e.g., increased water vapor content)
- Evidence of rapid ocean temperature changes

Triggering mechanism signatures:

- Spatial correlation between basin failures and volcanic/impact features
- Seismic modeling of basin response to external perturbations

- Pore pressure evolution models for metastable basin systems

These tests would not prove the model but could falsify specific predictions or provide support for the proposed mechanisms.

C.8. Future Work

The following areas require quantitative development before this system-level framework can be fully evaluated:

Critical gaps:

1. **Integrated thermal budget** including all potential heat sources (triggers) and sinks (atmospheric + crustal)
2. **Coupled ocean-atmosphere modeling** under globally inundated boundary conditions with specified heat input rates
3. **Advective cooling models** for large dilatant shear zones with realistic permeability and flow rates

Important refinements:

4. **Radiometric disturbance predictions** from catastrophic fluid flux through isotopic systems
5. **Sensitivity analysis** of various trigger scenarios on basin failure timing and spatial distribution
6. **Storm formation and persistence modeling** under anomalous thermal conditions

Strengthening additions:

7. **Coupled crust-hydrosphere simulations** to test cascading basin failure patterns
8. **Facies modeling** demonstrating hydraulic sorting produces observed stratigraphic patterns
9. **Modern analog studies** validating atmospheric heat export scaling relationships

These represent directions for continued research rather than demonstrated components of the present model. The core hydraulic collapse mechanism presented in the main paper stands independently of these extended considerations.

C.9. Summary

This appendix outlines a potential Earth-system context for hydrotectonic collapse: one in which hydrology, atmospheric circulation, and possible triggering events interact with a metastable crust already near failure.

Key findings:

1. **Atmospheric heat export** provides $\sim 60\times$ additional capacity beyond primary water-film dissipation during peak thermal stress (evaporative flux $\sim 100\text{--}400\text{ W/m}^2$, storm transport $\sim 10^{15}\text{--}10^{16}\text{ W}$)
2. **Triggering mechanisms** (supervolcanoes, impacts, progressive overpressure) are possible but not essential contributors to basin failure
3. **Surface hydrology** interacts with crustal deformation, potentially enhancing heat removal and affecting failure propagation
4. **Local vs. global perspective** distinguishes between observer experience (hydrological) and geophysical processes (crustal collapse)

These scenarios are offered as conceptual extensions to be tested, quantified, or falsified in future work. The core model presented in the main paper stands independently. The atmospheric calculations demonstrate that additional heat export mechanisms exist beyond the conservative primary estimate, providing margin against thermal runaway even under unfavorable assumptions.

References

Longmire, J.D. (2025). Rapid Continental Reorganization Through Hydraulic Collapse: A Solution to the Heat Problem in Catastrophic Plate Tectonics. *Zenodo*. <https://doi.org/10.5281/zenodo.17663309>

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